

Questions on Autoregressive Models

What are autoregressive models? (What is the autoregressive property?)

How can histograms be understood as generative models? What is the disadvantage?

What do we mean by parametric function approximation in this context? (keywords: Parametric distributions, MLE)

What is the KL-divergence measure? What is it used for?

Which kind of parametric models are we interested in?

What is the problem when modeling conditionals for high dimensional data?

How can we increase the efficiency (keyword: parameter sharing)?

How are parameters shared in RNNs (recap)?

How are parameters shared in CNNs (recap)?

How does PixelRNN work?

What is Wavenet? (What are dilated convolutions and how are they used in wavenet?)

How does PixelCNN work? (mention autoencoders here, explain why masking is needed)

What is the advantage of PixelCNN over PixelRNN?

What is the disadvantage?

Describe Gated PixelCNN? What is different from PixelCNN in terms of the receptive field?

What are the advantages of using attention in this context (name at least 3)?

How can class conditioning be realized in such models?